



Physical Activity and Its Relationship to BMI Among U.S. Adults

Introduction:

Obesity is a major public health issue in the United States and is linked to increased risk of chronic diseases. This project measures Body Mass Index (BMI) as it is used to assess weight status and overall health and Physical activity, because it is an important factor in maintaining a healthy body weight.

Research Question:

Is physical activity associated with BMI among U.S. adults?

Hypotheses

- H_0 : There is no difference in BMI between physically active and inactive adults
- H_a : There is a difference in BMI between physically active and inactive adults

Public Health Significance:

- Understanding the relationship between physical activity and BMI can help guide obesity prevention strategies.
- Findings support interventions promoting physical activity to improve population health.

Sample

U.S. adults with complete data after cleaning (complete-case analysis)

Variables

- BMI (Body Mass Index)
- Physical Activity (Yes/No)
- Age
- Sex
- Smoking status

Key Findings

- BMI values varied widely across participants
- Most individuals had BMI in the overweight/obese range
- Physically active adults generally had lower BMI

Interpretation of Graphs: The graphs shows that physically active individuals tend to have lower median BMI compared to inactive individuals.

Data and Methods

This study uses data from the National Health and Nutrition Examination Survey (NHANES), a nationally representative dataset collected by the Centers for Disease Control and Prevention (CDC).

Results

Descriptive analysis showed variation in BMI across physical activity levels. Visualizations indicated clear differences in BMI distributions between groups.

The ANOVA test revealed a statistically significant difference in BMI across physical activity levels:

- $F = 123$
- $p < 0.001$

These results indicate that BMI differs significantly depending on physical activity level.

